COURIER MANAGEMENT SYSTEM

User Story:

Courier companies face difficulties in managing shipments and deliveries due to manual processes and disconnected systems. Lack of centralized tracking causes delayed updates and poor customer visibility. Manual agent assignment and branch coordination can lead to operational delays. The absence of automation and performance analytics also reduces efficiency and customer satisfaction.

1. PROJECT OVERVIEW

The Courier Management System (CMS) is a Salesforce-based application designed to centralize courier operations. It manages customers, shipments, delivery agents, branches, delivery updates, invoices, reports, and dashboards in one platform.

1.1 Problem Statement

Manual shipment tracking and disconnected records can cause delayed updates, data errors, and difficulty in monitoring deliveries. The system addresses these issues through centralized data management and Salesforce automation.

1.2 Objectives

- Manage customer and shipment information.
- Assign delivery agents and maintain branch details.
- Track shipment status through delivery updates.
- Manage invoice and payment information.
- Use automation to update shipment status.
- Generate operational reports and dashboards.
- Provide role-based access to users.

1.3 Customer Journey

Booking → Shipment Creation → Agent Assignment → Transit → Delivery Update → Invoice → Report.

2. REQUIREMENT ANALYSIS

2.1 Functional Requirements

No.	Requirement
FR-1	Customer management
FR-2	Shipment booking

FR-3	Delivery agent assignment
FR-4	Real-time status updates
FR-5	Invoice generation
FR-6	Reports and dashboards

2.2 Non-Functional Requirements

- Usability – simple Salesforce Lightning interface.
- Security – role-based access control.
- Reliability – consistent automation and validation.
- Performance – fast tracking updates.
- Availability – cloud-based access.
- Scalability – supports multiple branches.

2.3 Data Flow

Customer Input → Shipment Creation → Validation → Agent Assignment → Delivery Update → Invoice → Reports.

3. TECHNOLOGY STACK

Layer	**Technology**
User Interface	Salesforce Lightning
Business Logic	Salesforce Flows
Database	Salesforce Objects
Automation	Validation Rules and Flows
Security	Profiles and Roles
Reporting	Reports and Dashboards

4. PROJECT DESIGN

4.1 Proposed Solution

CMS centralizes courier operations in Salesforce by connecting shipment management, delivery assignment, status tracking, billing, and reporting.

4.2 Solution Architecture

- Customer Object
- Shipment Object
- Delivery Agent Object
- Branch Object
- Delivery Update Object
- Invoice Object

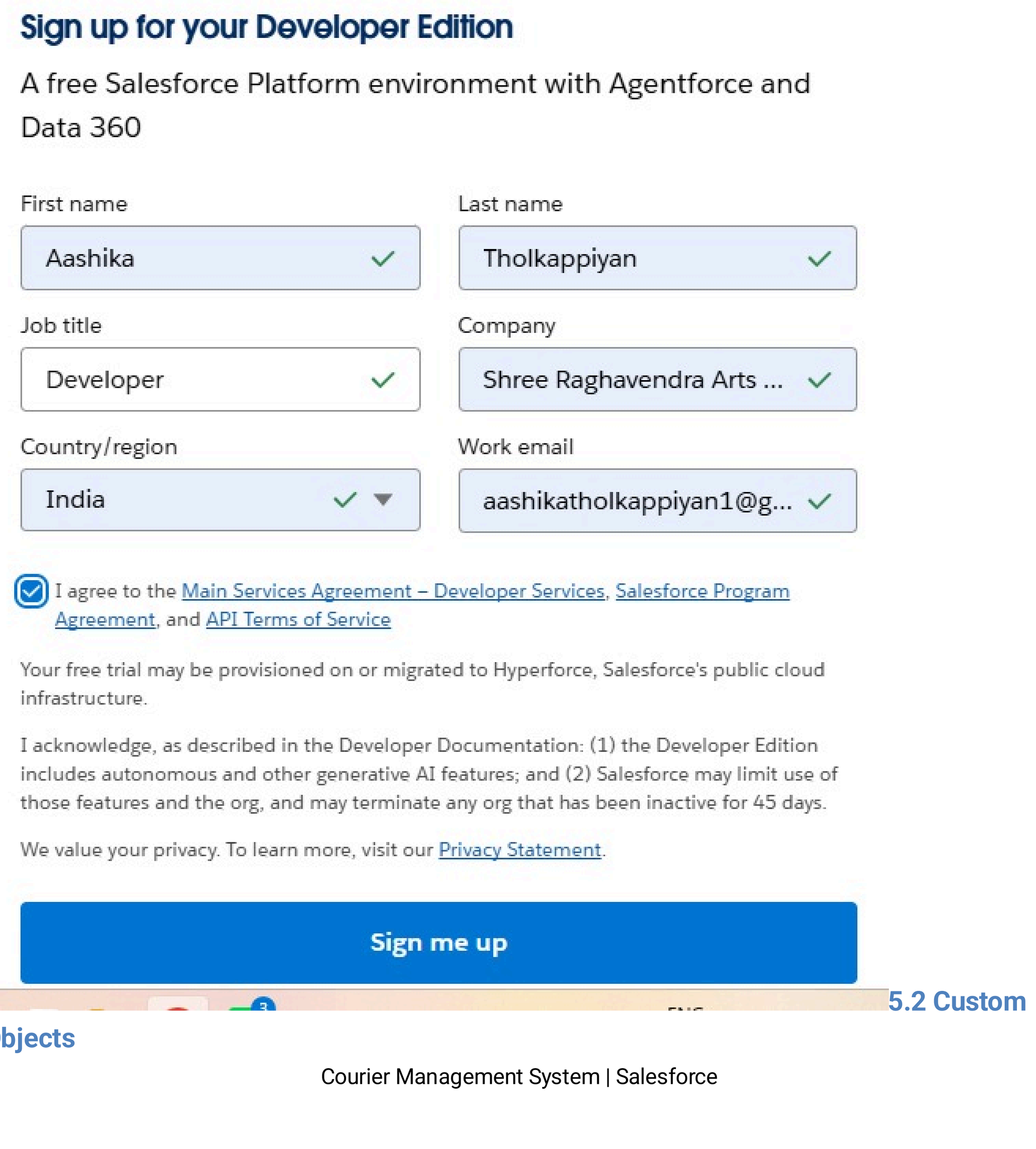
- Flows
- Reports
- Dashboards

5. SALESFORCE ACCOUNT AND DATA MODEL

5.1 Developer Account

A Salesforce Developer Edition account was created to configure and develop the Courier Management System.

Salesf



5.2 Custom Objects

Object	Purpose
Customer__c	Stores customer personal and contact information
Shipment__c	Stores parcel, source, destination and tracking information
Delivery_Agent__c	Stores delivery agent and branch details
Branch__c	Represents courier branch offices
Delivery_Update__c	Stores shipment status updates
Invoice__c	Manages billing information

5.3 Relationships

Child	Parent	Relationship
Shipment	Customer	Lookup
Shipment	Delivery Agent	Lookup
Delivery Agent	Branch	Lookup
Delivery Update	Shipment	Master-Detail
Invoice	Shipment	Lookup

6. KEY FIELDS

6.1 Shipment Fields

- Tracking Number
- Customer
- Shipment Type – Domestic / International
- Weight
- Price
- Status – Booked / In Transit / Out for Delivery / Delivered / Cancelled
- Expected Date
- Assigned Agent
- Source Address and Destination Address

6.2 Delivery Agent Fields

- Agent Name
- Email
- Phone
- Branch
- Status – Active / Inactive / On Leave
- Joining Date

6.3 Branch Fields

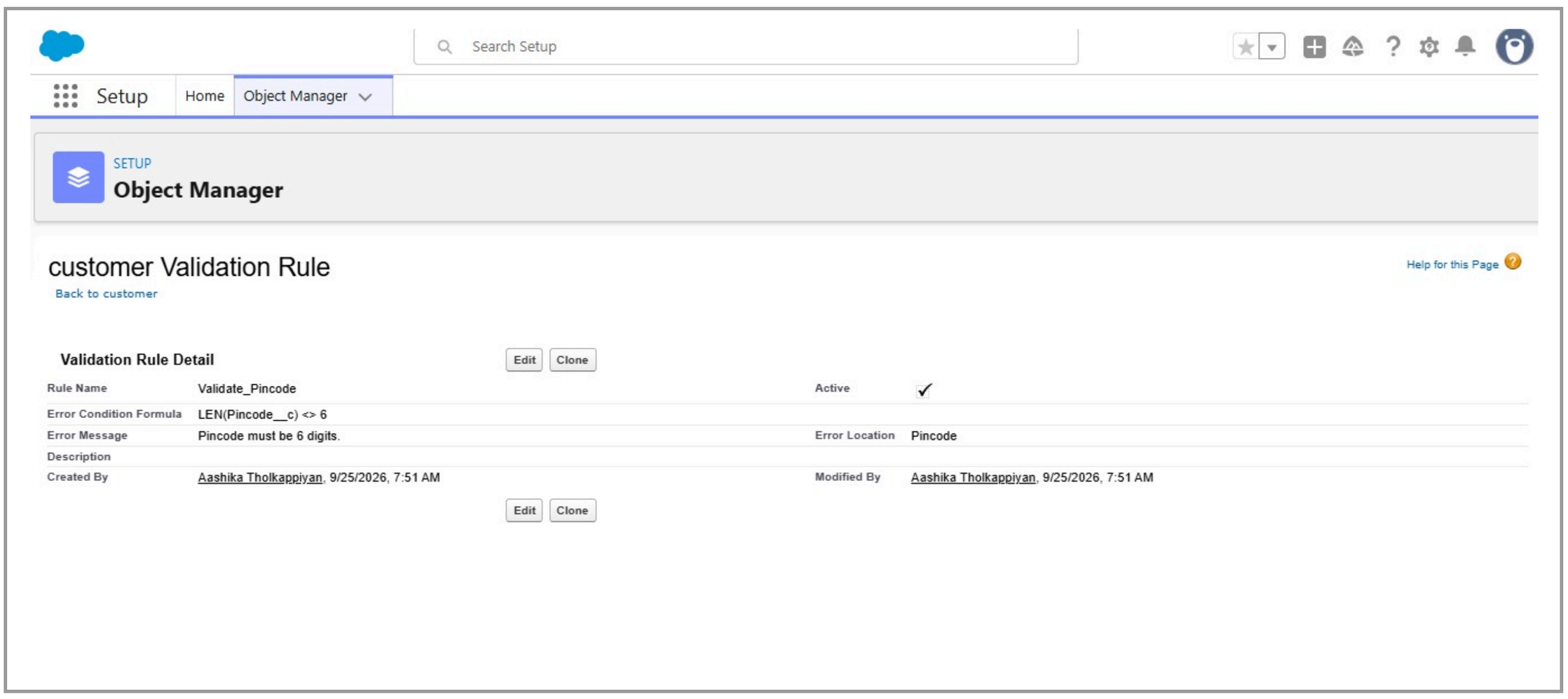
- Branch Name
- Branch Code
- City
- State
- Manager Name

6.4 Delivery Update Fields

- Update Number
- Shipment
- Status
- Update Date
- Remarks

6.5 Invoice Fields

- Invoice Number
- Shipment
- Amount
- Invoice Date
- Payment Status – Paid / Unpaid / Pending



7. VALIDATION RULES AND PAGE LAYOUTS

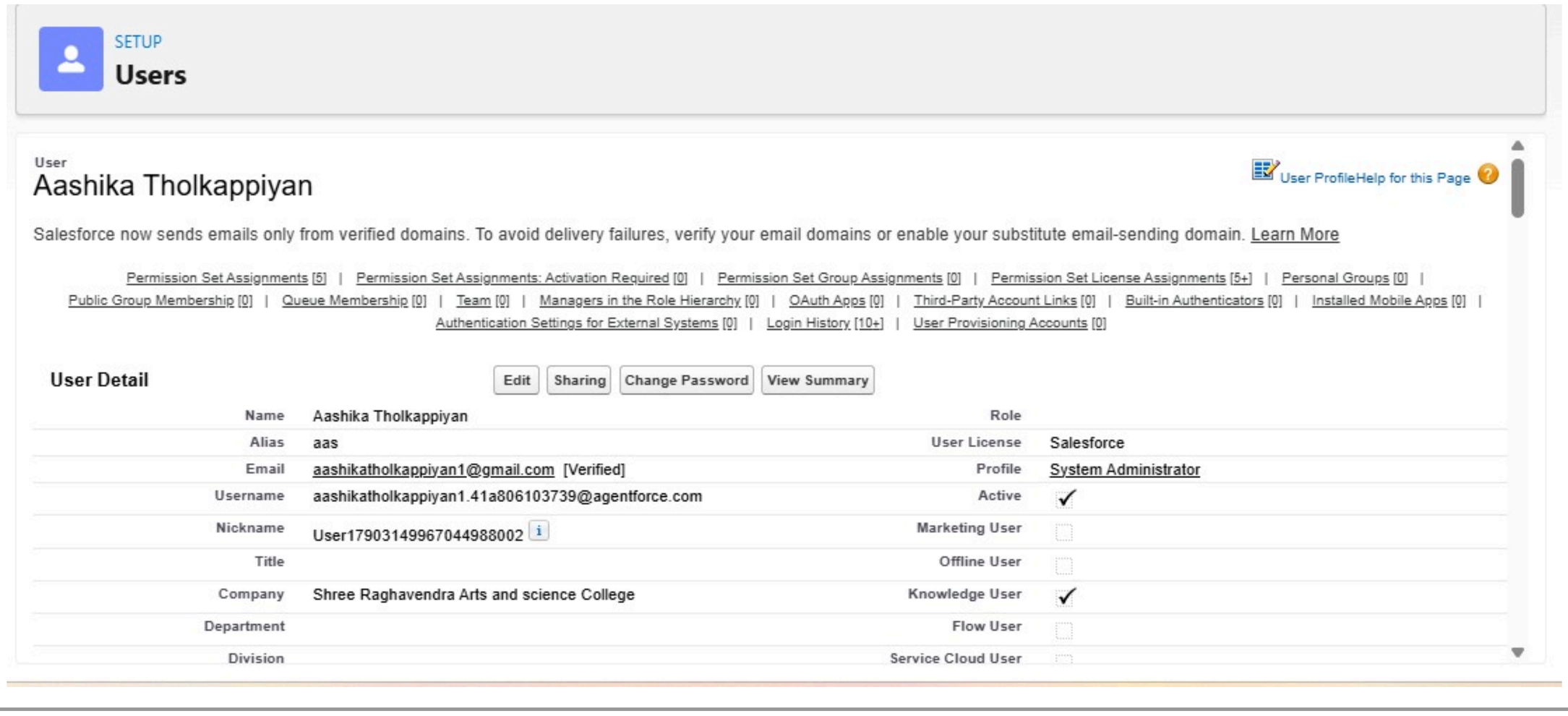
Validation rules are used to prevent incomplete or incorrect records. For example, required branch information and remarks for delayed shipments are validated before saving records.

Page layouts organize fields into sections so that users can easily enter and view required data.

Main Layout Sections

- Customer Information and Address Details
- Shipment Details, Address Information and Delivery Information
- Agent Details and Additional Information
- Branch Information

- Delivery Update Information
- Invoice Details

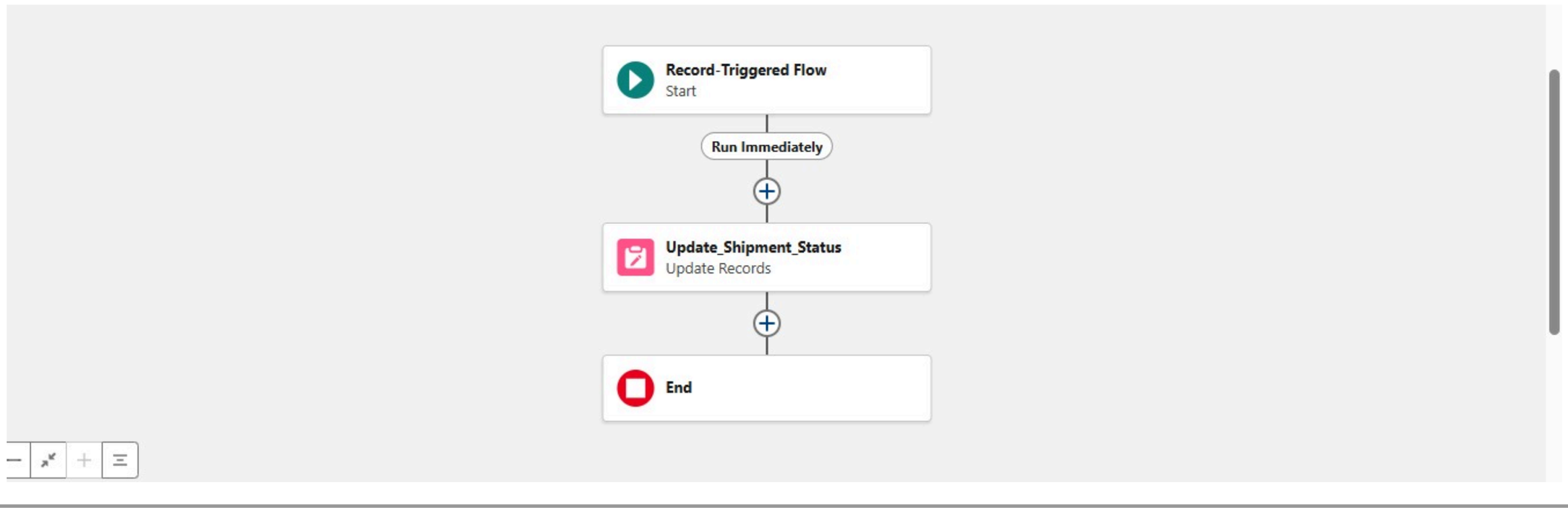


8. FLOW AUTOMATION

A record-triggered Flow is used to automatically update the related Shipment status when a Delivery Update record is created.

- Trigger: Delivery_Update__c record is created.
- Find the related Shipment record.
- Copy the Delivery Update status to Shipment Status.
- Save and activate the Flow.

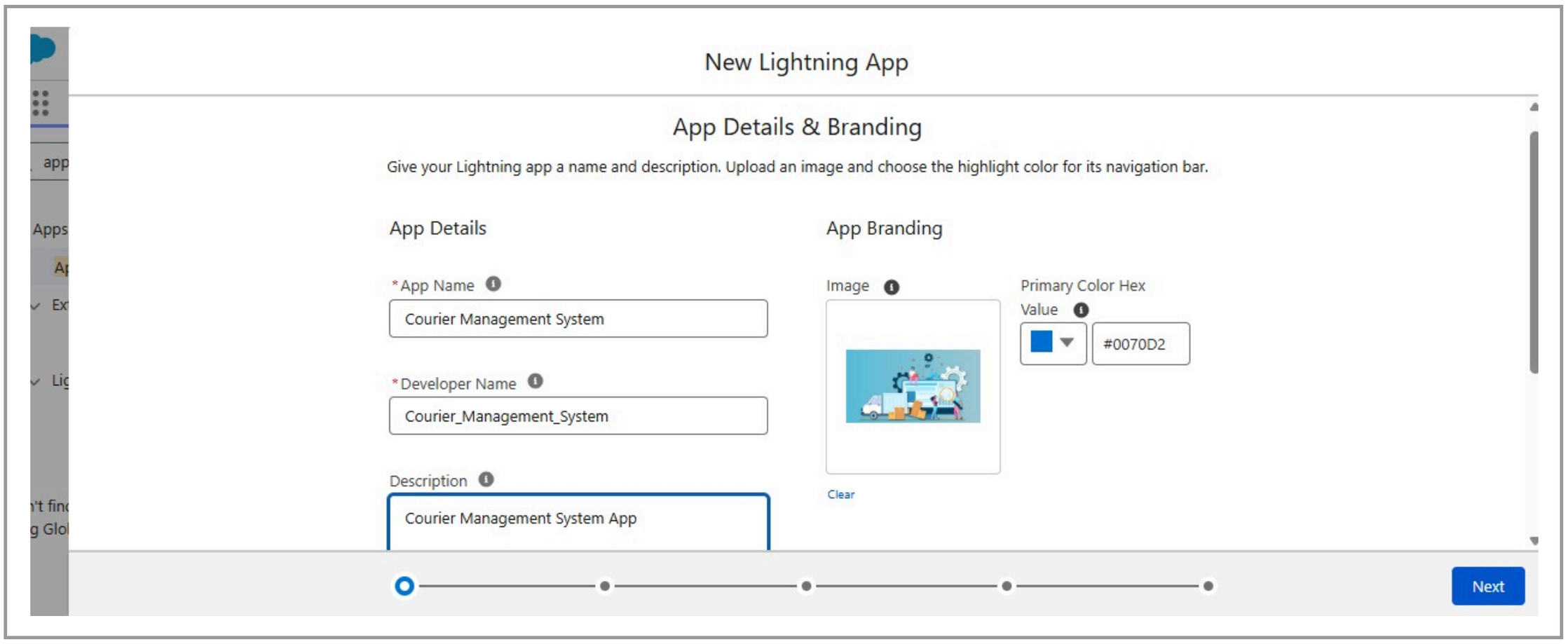
Flow Name: Auto_Update_Shipment_Status



9. LIGHTNING APP

The Courier Management System Lightning App groups the main courier modules into a single application for easier navigation.

- Customer
- Shipment
- Delivery Agent
- Branch
- Delivery Update
- Invoice
- Reports
- Dashboards



10. PROFILES, ROLES AND USERS

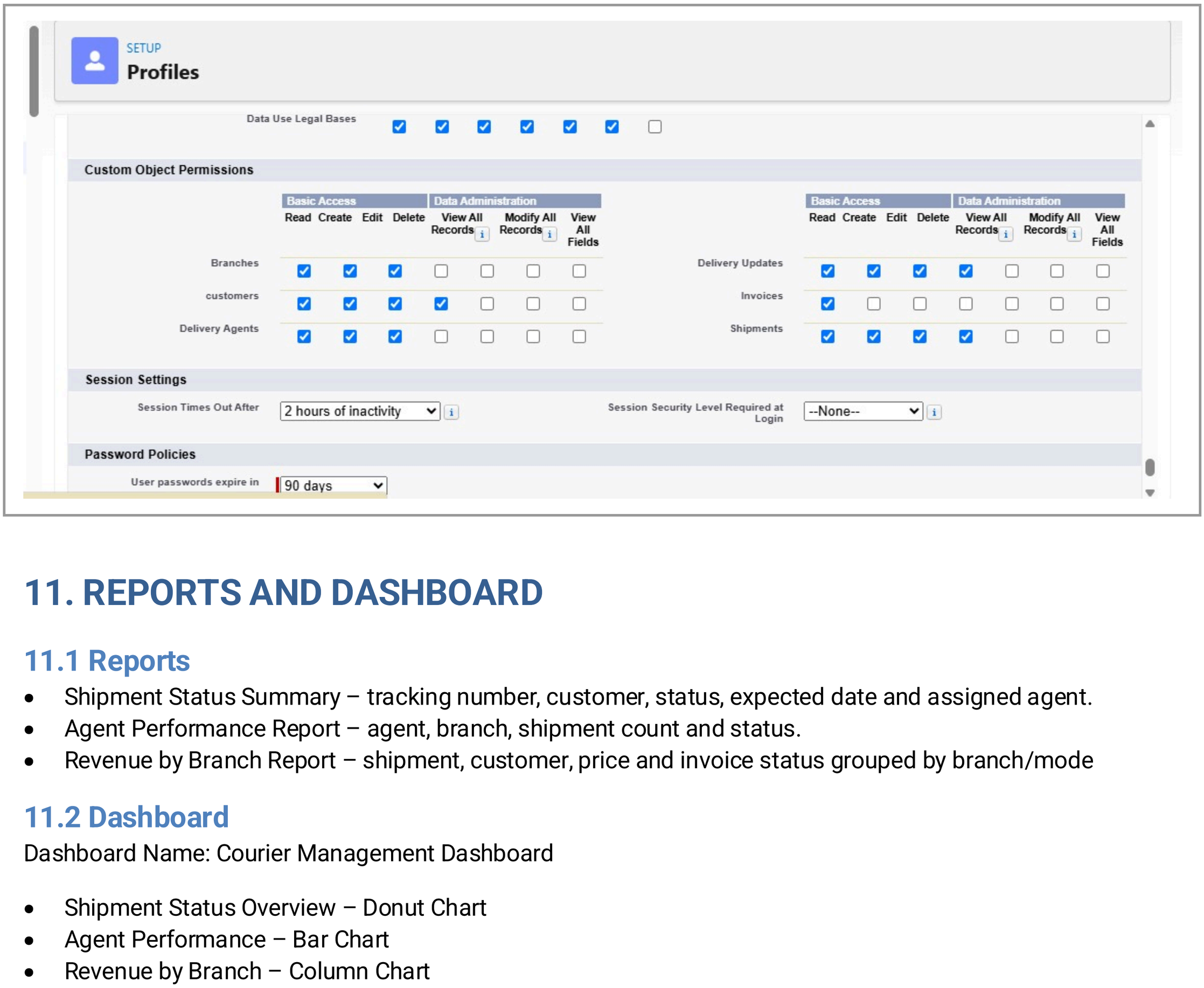
10.1 Profiles

Profile	Purpose
Courier Admin	Full system control
Branch Manager	Manage shipments and agents
Delivery Agent	Update delivery status
Customer Support	Handle customers and tracking

10.2 Role Hierarchy

Courier Admin → Branch Manager → Delivery Agent / Customer Support

Roles define the reporting hierarchy and help control access to organizational data.



11. REPORTS AND DASHBOARD

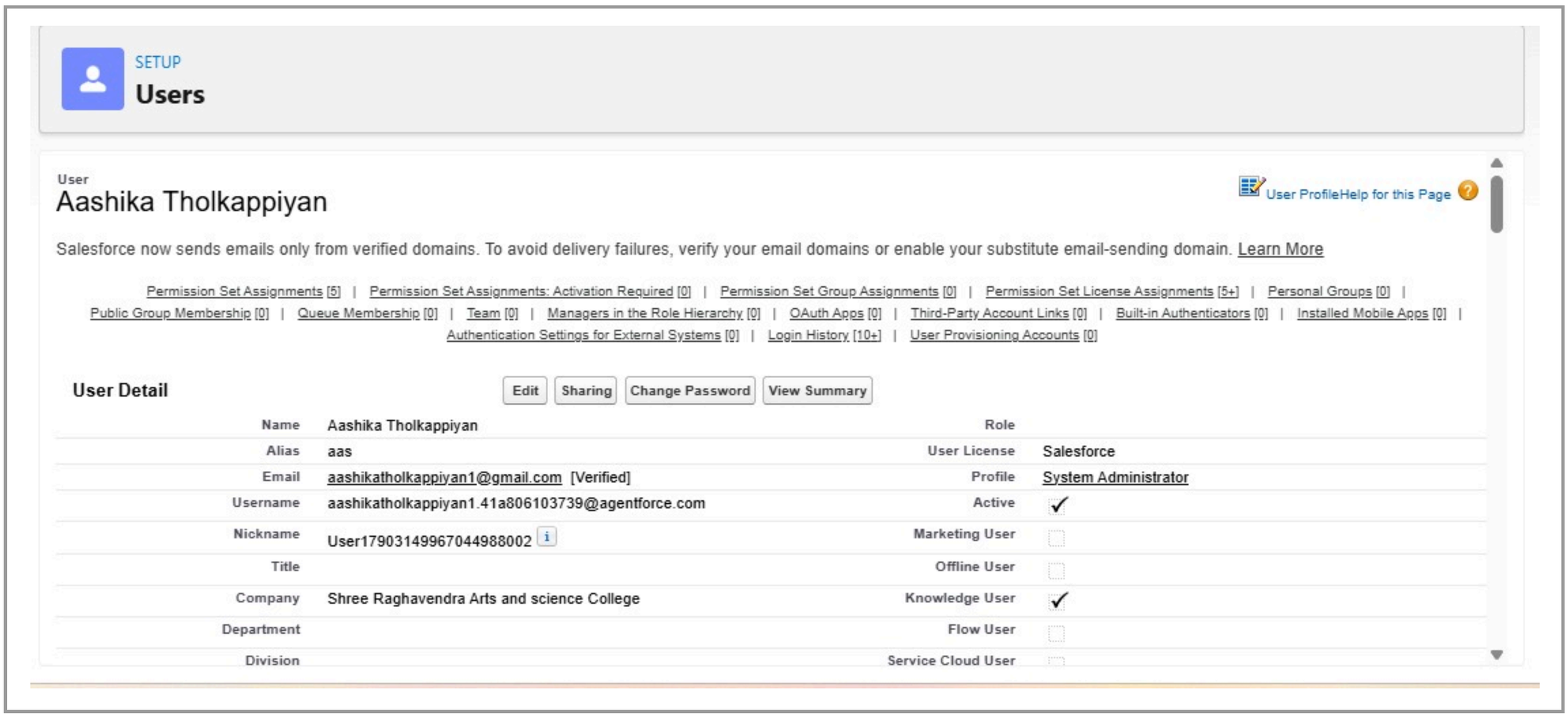
11.1 Reports

- Shipment Status Summary – tracking number, customer, status, expected date and assigned agent.
- Agent Performance Report – agent, branch, shipment count and status.
- Revenue by Branch Report – shipment, customer, price and invoice status grouped by branch/mode

11.2 Dashboard

Dashboard Name: Courier Management Dashboard

- Shipment Status Overview – Donut Chart
- Agent Performance – Bar Chart
- Revenue by Branch – Column Chart



12. FUNCTIONAL AND PERFORMANCE TESTING

Testing was performed to verify record creation, validation rules, delivery status updates, Flow automation, invoice information, and dashboard visualization. The configured scenarios were checked for correct operation and reliability.

Final Layout :



13. ADVANTAGES AND LIMITATIONS

Advantages

- Centralized data management
- Automated workflows
- Real-time reporting

Limitations

- Salesforce dependency
- Initial configuration effort

14. FUTURE SCOPE

- Mobile tracking application
- GPS integration
- SMS notifications

- AI-based delivery optimization

15. CONCLUSION

The Courier Management System digitizes courier operations using Salesforce. The system combines centralized data, automation, role-based access, shipment tracking, billing, reports, and dashboards to support efficient courier management.

16. APPENDIX

- Source Code: Salesforce Configuration
- Dataset: Salesforce Objects
- Demo: Salesforce Developer Org
- Screenshots: Project implementation screenshots